

# Diagnosing Transformation-Order and Inverse-Domain Errors

## Answer Key

Precalculus — Advanced function transformations and composition

### Quick Answers

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|------------------------------|--------------------------|
| 1. $x^2 - 6x + 11$           | 5. $2x + 8, -4, 0$       |
| 2. $4(x - 3)^2, 16$          | 6. $(x - 1)^2 + 4, 8$    |
| 3. $2x^2 - 5, 4x^2 + 4x - 2$ | 7. $(x - 3)^2 - 4, x, 6$ |
| 4. $\sqrt{3x - 1}, 2.236$    | 8. —                     |
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### Curriculum

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**Level:** Precalculus — Advanced function transformations and composition

HSF-BF.A.1c, HSF-BF.B.4 – *problems 3, 4, 6, 7, 8*

HSF-IF.A–HSF-IF.C – *problems 1, 2, 5*

*Difficulty 1–4 of 5 across 16 tagged checks.*

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### Common wrong answers

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- If they got 3.196: applied  $f$  first and  $g$  second, computing  $g(f(2))$  instead of  $f(g(2))$
  - If they got -8: shifted left 8 instead of left 4, so the tested input was -8
  - If they got 0: took the minus branch of the square root, which lands outside the restricted domain
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**1.**  $f(x) = x^2, g(x) = f(x - 3) + 2$ .

**Solution:** Substituting into  $f$  gives  $g(x) = (x - 3)^2 + 2$ , and expanding:  $x^2 - 6x + 9 + 2$ .

$$x^2 - 6x + 11$$

In words: the graph shifts right 3 (an inside operation, so it moves opposite to the sign) and up 2 (an outside operation, which moves the way it reads). This is the one case where inside and outside can be described independently, because there is only one of each.

**2.**  $h(x) = f(2x - 6)$  **with**  $f(x) = x^2$ .

**Solution (a):** factor the inside before reading anything off it:  $2x - 6 = 2(x - 3)$ , so  $h(x) = (2(x - 3))^2$ .

$$4(x - 3)^2$$

The horizontal transformations are: compress by a factor of  $\frac{1}{2}$  toward the  $y$ -axis, then shift right 3. Reading the unfactored inside would have given a shift of 6, which is twice the truth — the compression acts on the shift as well.

**Solution (b):**  $h(5) = (2(5) - 6)^2 = 4^2$ .

$$16$$

**3.**  $f(x) = 2x + 1$ ,  $g(x) = x^2 - 3$ : **both composites.**

**Solution:**  $(f \circ g)(x)$  means  $g$  runs first and its output is fed to  $f$ :

$$f(g(x)) = 2(x^2 - 3) + 1 = 2x^2 - 6 + 1$$

$$2x^2 - 5$$

Reversing the order feeds  $f$ 's output to  $g$ :

$$g(f(x)) = (2x + 1)^2 - 3 = 4x^2 + 4x + 1 - 3$$

$$4x^2 + 4x - 2$$

They differ because composition is not commutative: the first function's output becomes the second's input, so which one is applied first changes what the second one receives. A single test value makes it concrete — at  $x = 1$  the first gives  $-3$  and the second gives  $6$ .

**4. Find and fix:**  $(f \circ g)(2)$  **with**  $f(x) = \sqrt{x + 1}$ ,  $g(x) = 3x - 2$ .

**The mistake:** the notation was read left to right. In  $f \circ g$  the function written on the *right* runs first, so  $g$  takes the input and  $f$  takes  $g$ 's output. The student computed  $g(f(2))$ .

**Correct work:** compose symbolically first, which makes the order visible:

$$(f \circ g)(x) = f(3x - 2) = \sqrt{(3x - 2) + 1}$$

$$\sqrt{3x - 1}$$

Now substitute:  $(f \circ g)(2) = \sqrt{3(2) - 1} = \sqrt{5}$ .

$$2.236$$

*Watch for:*  $3.196$ , which is  $3\sqrt{3} - 2$  — the value of  $g(f(2))$ . The two composites are different functions, and here they are not even close; a symbolic composition before any arithmetic is what prevents the swap.

**5. Find and fix:** the graph of  $y = f(2x + 8)$ .

**The mistake:** the shift was read off the unfactored inside. A horizontal shift can only be read once the coefficient of  $x$  has been factored out, and "shift then compress" also has the order backwards.

**Correct work:** factor the inside:

$$2x + 8 = 2(x + 4)$$

So  $y = f(2(x + 4))$ : compress horizontally by a factor of  $\frac{1}{2}$ , then shift left  $4$  — not  $8$ . The compression halves the shift, which is exactly the trap in reading  $8$  off the constant term.

The inside is zero when  $2x + 8 = 0$ , that is at  $x = -4$ , confirming the shift: whatever feature of  $f$  sat at input  $0$  now sits at  $x = -4$ . Substituting that value back gives  $2(-4) + 8 = 0$ , as it must.

*Watch for:* testing  $x = -8$  instead. That input gives an inside value of  $-8$ , nowhere near the  $0$  the check is looking for.

**6.**  $f(x) = \sqrt{x-4} + 1$ : **inverse and its domain.**

**Solution (a):** write  $y = \sqrt{x-4} + 1$ , swap the variables, and solve for  $y$ :

$$x = \sqrt{y-4} + 1 \Rightarrow x-1 = \sqrt{y-4} \Rightarrow (x-1)^2 = y-4$$

Expanded this is  $x^2 - 2x + 5$ , and in completed-square form

$$(x-1)^2 + 4$$

**Solution (b):** the domain of  $f^{-1}$  is the range of  $f$ . Since  $\sqrt{x-4} \geq 0$ , the smallest output of  $f$  is 1, so the range of  $f$  is  $[1, \infty)$  and the domain of  $f^{-1}$  is  $x \geq 1$ . The algebra alone would happily accept any  $x$ , because  $(x-1)^2 + 4$  is defined everywhere; the restriction comes from the squaring step, which was only reversible because  $x-1$  was known to be non-negative.

**Solution (c):**  $f^{-1}(3) = (3-1)^2 + 4 = 4 + 4$ .

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Check the other way:  $f(8) = \sqrt{8-4} + 1 = 2 + 1 = 3$ .

**7. Find and fix:**  $f(x) = x^2 - 6x + 5$  on  $x \geq 3$ .

**The mistake:** two things at once. A  $\pm$  cannot appear in a function — it names two outputs for one input — and "domain: all real numbers" ignores that the domain of an inverse is the range of the original.

**Correct work:** complete the square to see the vertex:

$$x^2 - 6x + 5 = (x^2 - 6x + 9) - 4$$

$$(x-3)^2 - 4$$

The restriction  $x \geq 3$  keeps the right half of the parabola, where  $f$  is increasing and one-to-one. Solving  $y = (x-3)^2 - 4$  for  $x$  gives  $x = 3 \pm \sqrt{y+4}$ , and the surviving branch is the one whose outputs satisfy  $x \geq 3$ : the plus. So  $f^{-1}(x) = 3 + \sqrt{x+4}$ .

The check that this really inverts  $f$  is a composition:  $((3 + \sqrt{x+4}) - 3)^2 - 4 = (x+4) - 4 = x$ .

The range of  $f$  on  $x \geq 3$  starts at the vertex value  $-4$  and increases, so the range is  $[-4, \infty)$  and the domain of  $f^{-1}$  is  $x \geq -4$  — exactly where  $\sqrt{x+4}$  is defined, which is the algebra agreeing with the graph.

**Check:**  $f^{-1}(5) = 3 + \sqrt{5+4} = 3 + 3$ .

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*Watch for:* 0, the value of the rejected branch  $3 - \sqrt{5+4}$ . It is a real number and it satisfies  $f(0) = 5$ , which is why it looks right — but  $0 < 3$ , so it is outside the restricted domain and is not the inverse's output.

**8. Explain: order inside, and the domain of an inverse.** (*Open response — flagged for manual review; a model answer follows.*)

**Model answer:** An expression written inside a function is evaluated before the function is applied, so the transformations it encodes reach the graph in the reverse of the order they are written. In  $f(3x - 12)$  the input is multiplied by 3 and then 12 is subtracted, so on the graph the compression by  $\frac{1}{3}$  happens first and the shift second. The shift is not 12: factoring gives  $3(x - 4)$ , so the graph moves right 4. Reading a shift off an unfactored inside is the single most common source of the error, and the sign flips as well, because the input must grow by 4 to produce the value the untransformed function had at 0. Outside operations behave the opposite way — they are applied in the order written and in the direction written — which is why  $f(x) + 2$  really does shift the graph up 2.

The domain of  $f^{-1}$  is the range of  $f$  because the inverse undoes  $f$ : its inputs are exactly the outputs  $f$  produced, and no others. The algebra usually cannot see this, since the formula obtained by swapping and solving is often defined on a larger set. For  $f(x) = \sqrt{x - 4} + 1$  the formula  $(x - 1)^2 + 4$  is defined for every real  $x$ , but  $f$  never outputs anything below 1, so  $f^{-1}$  is only defined for  $x \geq 1$ . When the original function is not one-to-one, a restriction has to be imposed first; that restriction is what makes an inverse exist at all, and it decides which branch of a square root survives.

*What a reviewer should look for:* the inside factored before a shift is read; inside (input) transformations distinguished from outside (output) ones, including the reversed order and reversed direction; and the statement that a restriction on the original is what makes the inverse a function, with the domain of  $f^{-1}$  identified as the range of  $f$ .